Bilingual Skill Routing at 80K Scale: Domain Adaptation and Instruction Prefixes

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Abstract

Agent skills are reusable instruction packages selected before an agent loads their full procedures. This progressive-disclosure design makes routing a high-leverage decision, but current skill-routing benchmarks are English-centric. We introduce a bilingual evaluation that adds faithful Chinese translations and fixed bilingual-instruction variants to 73 public SkillRouter tasks while preserving their expert skill labels. We retrieve from 79,141 hard-tier skills using word TF–IDF, BM25, character TF–IDF, multilingual-E5, a skill-domain encoder, and reciprocal-rank fusion. The domain encoder obtains 41.1% Hit@1 on Chinese queries versus 8.2% for word TF–IDF. Complementing it with character evidence reaches 42.5% Hit@1 in Chinese and 39.7% when a fixed bilingual instruction prefixes the Chinese task. The results expose two practical asymmetries: domain adaptation matters more than generic multilinguality, and a repeated bilingual instruction can reduce dense retrieval quality even when it helps lexical fusion. All evaluations are local and reproducible; we release query variants, checksums, code, confidence intervals, rankings, and latency measurements.

Introduction

Agent hosts increasingly package reusable behavior as *skills*: folders containing a short name and description, a full instruction file, and optional scripts or references. The Agent Skills specification formalizes this progressive-disclosure pattern: metadata is always visible, instructions are loaded after activation, and larger resources are read only when needed (Agent Skills 2025; Zhang, Lazuka, and Murag 2025). This design saves context, but it also creates an upstream bottleneck. If the router surfaces the wrong skill, a capable downstream model receives the wrong procedure.

Recent work establishes that the bottleneck is real. SkillsBench measures whether skills improve task execution (Li et al. 2026), while SkillRouter evaluates retrieval from roughly 80K skills and finds that full skill text provides important evidence in highly overlapping pools (Zheng et al. 2026). These benchmarks, however, are distributed as English data. A common deployment setting is bilingual: a user states intent in Chinese while the registry metadata remains entirely English. Hosts may also prepend a bilingual routing instruction to a non-English request. This cross-lingual mismatch occurs *before* tool execution, so evidence about multilingual function calling (Guo, Huang, and Xiong 2024; Luo et al. 2026) does not tell us whether the correct skill is discovered.

We study this problem by constructing Chinese and instruction-prefixed Chinese variants of the public SkillRouter hard-tier tasks and retrieve from 79,141 candidates. The variants preserve the original gold skills and all executable constraints. We compare sparse lexical methods, a compact generic multilingual encoder, the released skill-domain encoder, and rank fusion.

Our contributions are:

- a 73-task, 219-query bilingual extension of a public 79K-skill benchmark, with English, Chinese, and fixed bilingual-instruction variants tied to the same expert labels;
- an evaluation of TF–IDF, BM25, character retrieval, generic multilingual-E5, a skill-domain encoder, and fusion under Hit@1, Hit@5, recall@10, NDCG@10, confidence intervals, paired tests, indexing cost, and query latency; and
- a task-level analysis showing when domain adaptation transfers across language and how a fixed bilingual instruction changes sparse, dense, and fused retrieval differently.

Related Work

Skills and large registries. The Agent Skills specification defines short routing metadata and delayed instruction loading (Agent Skills 2025). SkillsBench evaluates the downstream value of modular procedures (Li et al. 2026). SkillRouter directly studies large-pool retrieval and introduces a learned retrieve–rerank pipeline over full skill text (Zheng et al. 2026). Our work keeps SkillRouter's hard candidate pool and expert labels, but asks a different question: how robust is *metadata-only* routing when queries change language?

Tool retrieval and awareness. ToolLLM retrieves from more than 16K APIs and trains models for tool use (Qin

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et al. 2024); AnyTool uses hierarchical retrieval and reflection (Du, Wei, and Zhang 2024). MetaTool explicitly separates tool-use awareness from tool selection (Huang et al. 2024). These systems route executable API schemas or plugins, whereas skills are procedural text whose full content may not be loaded until after selection.

Multilingual retrieval and execution. CToolEval evaluates Chinese API interaction (Guo, Huang, and Xiong 2024), and Lost in Execution analyzes multilingual tool-calling failures (Luo et al. 2026). MIRACL provides multilingual document retrieval (Zhang et al. 2023), while multilingual-E5 supplies compact cross-lingual embeddings (Wang et al. 2024). We connect these threads by evaluating multilingual retrieval over procedural skill metadata rather than documents or post-selection API calls.

Efficient retrieval and fusion. We use classical term weighting (Salton and Buckley 1988), BM25 (Robertson and Zaragoza 2009), and reciprocal-rank fusion (Cormack, Clarke, and Büttcher 2009). Unlike learned rerankers, these methods make it possible to isolate lexical overlap, domain adaptation, and query latency.

Task and Data

Large-Pool Cross-Lingual Retrieval

We use the public SkillRouter hard tier, which contains 79,141 skill records. Each record has a stable identifier, name, description, body, and source. We route using only `name + description`; this matches the metadata available before progressive disclosure and is deliberately weaker than SkillRouter's full-body setting. The source benchmark provides 87 tasks. We follow its evaluation rule by removing two source-excluded tasks and 12 `generic_only` tasks, leaving 73 tasks with 182 distinct core gold skills. Every core gold identifier is present in the hard pool.

For each English task, we add:

1. a faithful Simplified Chinese translation that preserves filenames, code symbols, schemas, numbers, and negations; and
2. the same Chinese task prefixed by one fixed bilingual instruction that asks the router to preserve technical identifiers and constraints.

This produces 219 queries. A language model generated the first pass, and a separate isolated run audited every row against the English source. Our validator checks exact English preservation, task coverage, JSON integrity, and the presence of both Latin and Chinese scripts in the instruction-prefixed variants. It additionally verifies that every such variant is exactly the fixed prefix plus its audited Chinese task. Gold labels are inherited unchanged from the source benchmark; no new relevance label is model-generated. We report this construction process because the translations are not human-authored ground truth.

Methods

Word TF–IDF. We concatenate name and description, lowercase Latin text, and fit unigram and bigram TF–IDF with sublinear term frequency and at most 300K features. Cosine similarity ranks all candidates (Salton and Buckley 1988).

BM25. We use unigram BM25 with $k_1 = 1.2$ and $b = 0.75$ (Robertson and Zaragoza 2009). Query terms are binary, while document term frequency is length-normalized. This baseline exposes the effect of lexical mismatch when Chinese intent must retrieve English metadata.

Character TF–IDF. Character 2–5-grams provide partial matching across identifiers, spelling variation, and bilingual instruction text. We keep features occurring in at least two documents and cap the vocabulary at 300K.

Multilingual-E5. We encode metadata with `intfloat/multilingual-e5-small`, a 118M parameter, 384-dimensional multilingual retriever (Wang et al. 2024). We use the recommended `passage:` and `query:` prefixes, normalized embeddings, a 512-token limit, and exact inner-product search. No task-specific training is performed.

SkillRouter encoder. We also evaluate `SkillRouter-Embedding-0.6B`, a 596M parameter Qwen3-based bi-encoder trained for skill retrieval (Zheng et al. 2026). We keep the study metadata-only by formatting each document as `name | description`, use the model's task-retrieval instruction, left padding, last-token pooling, normalized 1024-dimensional embeddings, and a 512-token limit. The model is trained on English skill routing but inherits a multilingual base model, making its cross-lingual transfer an empirical question.

Fusion. We combine the top 20 character and SkillRouter-encoder rankings using reciprocal-rank fusion with constant 60 (Cormack, Clarke, and Büttcher 2009). This tests whether lexical identifier matches complement domain-adapted semantic retrieval.

Metrics and statistics. Hit@1 and Hit@5 indicate whether any core gold skill appears by rank 1 or 5. Recall@10 measures the fraction of all core gold skills recovered. NDCG@10 uses the source benchmark's graded relevance, including degraded variants. MRR@20 uses the first core gold within the saved top 20 and assigns zero otherwise. NDCG's ideal ranking includes only judgments present in the hard candidate pool. All score ties use ascending skill identifier as a stable secondary key. We compute 95% task-bootstrap confidence intervals with 10,000 samples and fixed seed 20270827. Exact two-sided McNemar tests compare paired Hit@1 outcomes. Latency includes query encoding and exact search but excludes setup. Sparse index fitting and dense cache loading plus validation are reported as `setup_seconds`; offline candidate embedding generation is not timed.

Results

Cross-Lingual Retrieval at 79K Scale

English lexical retrieval is competitive because queries and metadata share language. The picture changes under translation. Word TF–IDF falls from 37.0% Hit@1 in English

	Hit@1			Recall@10			
Method	EN	ZH	ZH+Inst	EN	ZH	ZH+Inst	ms/query
Word TF–IDF	37.0%	8.2%	8.2%	46.8%	17.7%	17.7%	5.7
BM25	46.6%	19.2%	12.3%	46.0%	21.7%	17.1%	5.6
Char TF–IDF	**49.3%**	21.9%	24.7%	50.1%	31.4%	29.4%	14.8
mE5-small	13.7%	0.0%	0.0%	15.5%	0.0%	0.0%	12.0
SR-Emb-0.6B	45.2%	41.1%	26.0%	50.5%	43.8%	38.9%	140.8
SR+Char RRF	**49.3%**	**42.5%**	**39.7%**	**54.9%**	**46.0%**	**42.1%**	155.6

Table 1: Metadata-only retrieval on the SkillRouter hard tier. All methods rank the same 79,141 English skill names and descriptions. Bold marks the best value in each quality column. Latency excludes one-time index construction.

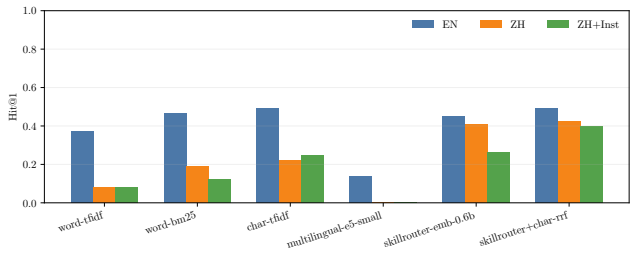


Figure 1: Hit@1 by query condition on the SkillRouter hard tier. ZH+Inst is the audited Chinese task with one fixed bilingual routing instruction. All conditions retrieve the same English metadata and use the original labels.

to 8.2% in Chinese. Generic multilingual-E5 reaches only 0.0%, showing that multilingual pretraining alone does not solve routing in a highly overlapping registry. The skill-domain encoder reaches 41.1%, and its paired comparison with character TF–IDF is $p = 0.013$, exact McNemar, 21 gains vs. 7 losses. Fusion is strongest at 42.5%.

The fixed bilingual instruction does not uniformly help. It leaves word TF–IDF unchanged at 8.2%, reduces BM25 from 19.2% to 12.3%, and reduces the domain encoder from 41.1% to 26.0%. Character retrieval rises slightly to 24.7%, and fusion recovers 39.7% Hit@1 and 42.1% recall@10. Thus a repeated bilingual instruction is not a substitute for task-specific cross-lingual evidence. Figure 1 supports reporting the two Chinese conditions separately rather than pooling them.

Latency. Sparse methods answer in 5.6–14.8 ms per query. E5 requires 12.0 ms per query after its candidate embeddings are cached. The domain encoder requires 140.8 ms and fusion requires 155.6 ms. The cached mE5 and domain matrices occupy 116 MB and 309 MB, respectively. These measurements are on Apple MPS and should not be interpreted as server throughput.

Analysis

Language mismatch is the main sparse failure mode. English metadata contains many exact API names and domain terms. Chinese translations preserve the task but remove most lexical overlap. The fixed instruction restores only a repeated set of English routing terms, not task-specific evidence. The domain encoder reduces the translation loss without translating 79K descriptions, which is useful when registries aggregate skills from many authors.

Domain adaptation dominates generic multilinguality. The generic multilingual-E5 model performs poorly even on English and retrieves no core gold at rank 1 for Chinese or instruction-prefixed queries. A sanity check confirms that it aligns short English and Chinese sentences, so the failure is not a broken language encoder. Rather, generic semantic similarity is insufficient to distinguish tens of thousands of overlapping procedural descriptions. Skill-routing training produces far stronger transfer even though the released checkpoint is labeled English. The generic encoder's most common Chinese top-1 prediction appears for 12 of 73 tasks; the domain encoder produces 71 distinct top-1 predictions and no candidate appears more than twice. Against character TF–IDF, the domain encoder gains 21 Chinese tasks and loses seven, but its instruction-prefixed comparison is nearly even (14 gains, 13 losses). RRF then gains 16 instruction-prefixed tasks over the domain encoder while losing six, showing that lexical evidence can recover task terms when a fixed bilingual instruction distracts the dense model.

Instruction prefixes are model dependent. The fixed prefix leaves word TF–IDF unchanged, hurts BM25 and the domain encoder, but improves character retrieval and makes fusion much stronger than the domain encoder alone. A host therefore cannot assume that an English or bilingual meta-instruction helps retrieval simply because it helps an instruction-following model downstream. Routing prompts and task prompts have different objectives and should be evaluated separately.

Limitations and Ethical Considerations

The multilingual variants are generated and audited by language models, not professional translators. Although executable constraints and gold labels are preserved, subtle phrasing errors may remain. The public benchmark has only 73 evaluated tasks, so confidence intervals are wide and domain slices are underpowered. We evaluate metadata only and truncate dense inputs to 512 tokens; results do not establish the behavior of full-body retrieval or longer-context encoders. Some candidate IDs have identical name–description text; a stable ID tie-break makes evaluation reproducible, but exact-ID metrics still penalize metadata-indistinguishable

duplicates. We do not measure end-to-end task success or no-skill abstention.

We use only public SkillRouter data. Raw private logs, credentials, work records, leaked source, and identifying author metadata are excluded. The released multilingual file contains task text from the source benchmark and must retain its upstream attribution and license terms. Automated routing can activate instructions with filesystem or shell permissions; higher retrieval accuracy does not remove the need for permission checks and provenance.

Conclusion

Bilingual skill routing is not only a smaller version of English retrieval. At 79K scale, multilingual dense embeddings recover skills lost to lexical translation, while a fixed bilingual instruction is not a reliable substitute for task-specific evidence. Future work should jointly optimize cross-lingual retrieval, prompt design, full-text routing, and downstream task success.

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